

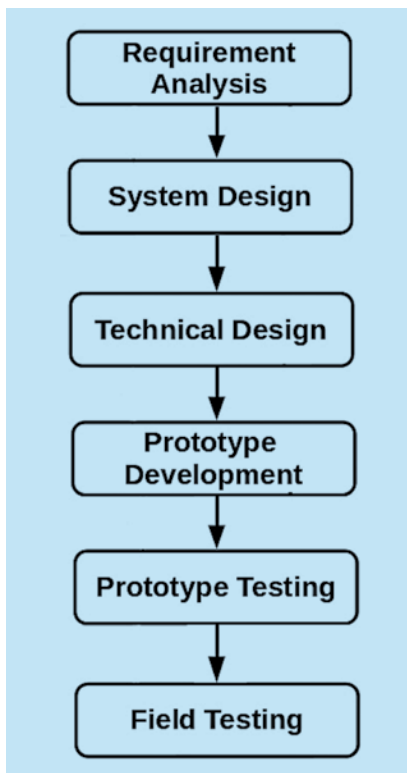


Fig. 1: Waterfall development model

helped to develop a draft product definition. This product definition needs further detailing and has to go through several iterations before we get our concepts cleared and develop a marketable product plan.

There can be various paths of product development. In technical language, we call this path a development model. Each of the product development models has certain advantages and certain drawbacks. The first decision we take in product development is to decide on this development model. Let us look into some of the popular development models and understand where to use which one.

Waterfall model

The traditional approach to product development had been to start with listing the requirements of the product. Once these requirements are frozen, then the product architecture is defined. Then the technical detail design is made, which is followed by an implementation plan, design



Fig. 2: Spiral development model

verification, maintenance plan, distribution plan, and marketing plan. This model is called the waterfall development model (see Fig. 1).

The waterfall model simplifies communication between different skill groups. In this model, each group works on the product concept at a time. Once a team finishes its work, it hands over the outcome of its work to the next team. In the days when communication facilities were in their infancy, computing resources were very costly, and waterfall model was the only option available to product developers. In those days, this model worked nicely.

The problem with waterfall model was the time taken to develop a product. Each team worked in a sequence, which resulted in a long delay between product conceptualisation and its realisation.

The second problem was the lack of interaction between different groups. This lack of interaction often resulted in manufacturing difficulty, poor quality, and maintainability. At times, such poor interactions resulted in costly changes in design and manufacturing processes at advanced stages of development.

The waterfall model is not suitable for developing complex products.

Concurrent engineering

These days we use a process called concurrent engineering (CE). In the CE approach, evaluation, designing, and planning happen together. Team members with different skills get involved in the designing process right from the beginning and evolve the product by working together.

Let me give you a small example of how the involvement of members from different functional groups helps to improve product design. This author was developing a control system of an automated press for making compressed earth blocks. In the

press, a mixture of soil and cement is compressed to form blocks. These blocks are an environment-friendly alternative to terracotta bricks.

Manufacturing of these blocks required close dimensional tolerances. As the ram presses the soil it has some inertia. For this reason, power to the ram needs to be cut off ahead of the endpoint. The distance the ram travels after power cut-off is governed by the plasticity of the soil. This property varies widely from batch to batch. Sensing the point where to cut off the power posed a challenge with conventional sensors.

During the prototype trial of the press, one of the production team members came up with the idea to fix a stopper at the endpoint. He suggested using the sound ram made when it hit that stopper as the sensing system for adjusting the endpoint sensor. Under normal conditions, the ram will touch the stopper with a soft click. If the clicking sound is missed, then he will lower the sensor. If the ram made a hard clapping sound, then he will raise the sensor. That operational innovation greatly simplified the design and improved dimensional quality of the bricks.

Current advances in communication and collaboration technologies